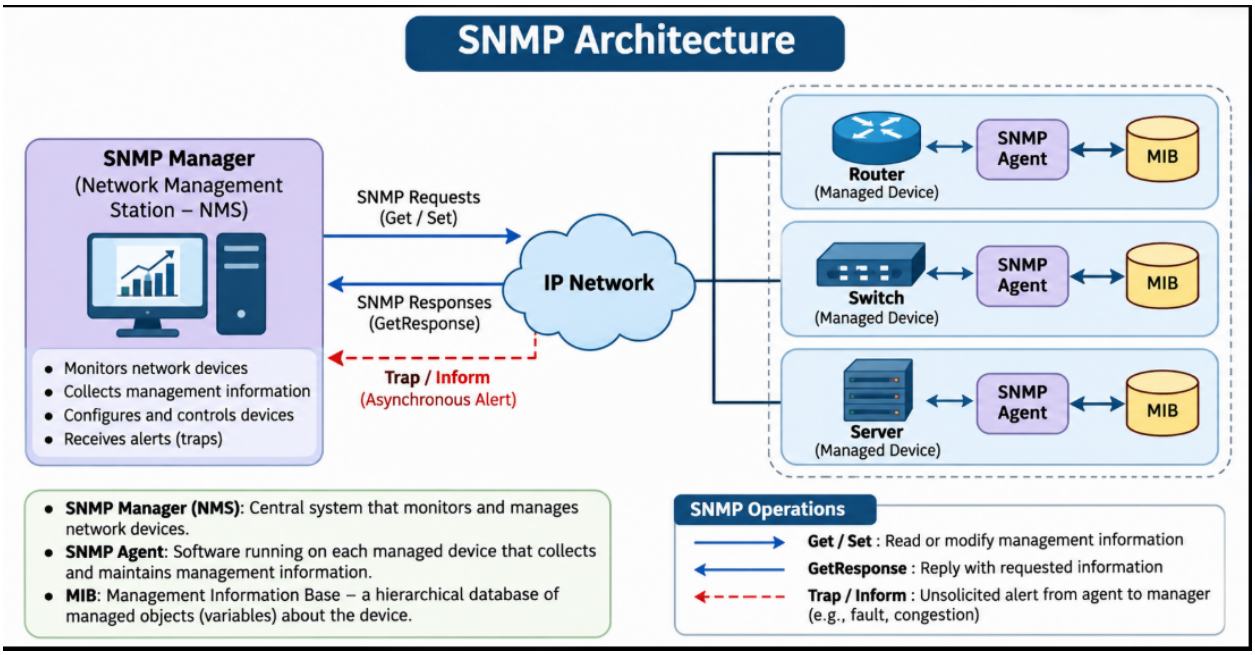
SNMP
(SIMPLE NETWORK MANAGEMENT PROTOCOL)

SNMP (Simple Network Management Protocol) is an application-layer protocol used for monitoring, managing, configuring, and troubleshooting network devices such as routers, switches, and servers.

It helps network administrators ensure the smooth and efficient operation of a network and maintain the required Quality of Service (QoS).

SNMP uses three main components:

1. SNMP Manager: Monitors and manages network devices.
2. SNMP Agent: Software running on network devices that collects and provides information to the manager.
3. Management Information Base (MIB): A database containing information about the managed network devices.



1. SNMP Manager

- It is a centralized system used to monitor and manage the network.
- It is also called a Network Management Station (NMS).
- It runs the SNMP client program and communicates with SNMP agents to collect network information.

2. SNMP Agent

- It is a software module installed on a managed network device, such as a router or switch.
- It collects and maintains information about the device in a database.
- The manager accesses this information to monitor network performance.
- For example, the manager can check the number of packets transmitted and received by a router to determine network congestion.

3. Management Information Base (MIB)

- MIB is a database that contains information about the resources to be managed in a network.
- The information is organized in a hierarchical structure.
- It consists of managed objects, which represent variables containing information about network devices.
- The SNMP agent maintains these objects, and the manager accesses them to monitor and manage the network.
- The major groups in MIB-II are System, Interfaces, Address Translation, IP, ICMP, TCP, UDP, and EGP.